

Azeotropic mixture

1. The azeotropic mixture of water ($b.p. 100^{\circ}C$) and HCl ($b.p. 85^{\circ}C$) boils at $108.5^{\circ}C$. When this mixture is distilled it is possible to obtain
 - (a) Pure HCl
 - (b) Pure water
 - (c) Pure water as well as pure HCl
 - (d) Neither HCl nor H_2O in their pure states
2. An azeotropic solution of two liquids has boiling point lower than either when it
 - (a) Shows a negative deviation from Raoult's law
 - (b) Shows no deviation from Raoult's law
 - (c) Shows positive deviation from Raoult's law
 - (d) Is saturated
3. A liquid mixture boils without changing constituent is called
 - (a) Stable structure complex
 - (b) Binary liquid mixture
 - (c) Zeotropic liquid mixture
 - (d) Azeotropic liquid mixture
4. Azeotropic mixture are
 - (a) Constant temperature boiling mixtures
 - (b) Those which boils at different temperatures
 - (c) Mixture of two solids
 - (d) None of the above
5. A mixture of two completely miscible non-ideal liquids which distil as such without change in its composition at a constant temperature as though it were a pure liquid. This mixture is known as
 - (a) Binary liquid mixture
 - (b) Azeotropic mixture
 - (c) Eutectic mixture
 - (d) Ideal mixture

